Master 1 Lab 1 : Solve the Jump game

We consider a board of size $n \times m$. The initial board has some configuration of colored square tokens with at least one empty space.

The objective is to remove all but one square token from the board.

colored square tokens are removed when they are jumped according to the following rules.

Rules: If a square token of one color jumps over a different colored square token to an empty space, the jumped square token is removed from the board. Additionally, if multiple square tokens of the same color are in a line, they can be jumped and removed together.

Objective: we want to find a solution if there is one. We also want to optimize either the complexity, either memory, or both.

The following figures show possible moves from the board. The outcome of the action on the left board is on its right.





